

Mostafa Kmal

AI & Informatics (Robotics) Student · Games, Apps & Hardware

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EDUCATION

Horus University — B.S. in Artificial Intelligence & Informatics (Robotics)

Sep 2025 – Present

Level 2 · Faculty of Artificial Intelligence · New Damietta, Egypt

AWARDS & COMPETITIONS

- 3rd Place — [Green Loop Competition](#), Horus University round with ReSpark — qualified to represent the university nationally. Sep 2026
- Rank #1 at Horus University — [ECPC 2026 Qualifications](#) with ICPC HUE — #189 on Day 7, 4 problems solved in 5 hours as first-year students. Aug 2026
- Qualified — [Damietta Hackathon](#) — passed the preliminaries and built a working ESP32 medical wearable on site. Jul 2026

FEATURED PROJECTS

[fosh&fish](#) — Cozy 3D Fishing Game, Godot 4 + Blender

Oct 2026

- Rebuilt the Virtual Fisher Discord bot as a full game: 20 fish, 21 rods, 17 boats, 8 baits and 7 biomes balanced against the bot's data; ships to Web, Windows, Linux and Android.
- Built a scripted Blender art pipeline, headless simulation tests for mechanics and balance, and an automated trailer pipeline; open source under MIT.

[ReSpark](#) — E-Waste Recycling Platform & Mobile App

Sep 2026

- Built the mobile application with a teammate for a platform where students resell parts of old hardware projects, plus software strategies that cut AI energy and water use.
- Won 3rd Place at Green Loop (HUE round) in a team of three; qualified for the national round.

[Smart Medical Watch](#) — ESP32 Wearable & Live Dashboard

Jul 2026

- XIAO ESP32-C6 watch with a 1.28" round TFT, MPU6050 step counting, MAX30100 pulse sensing (stabilised with 4.7kΩ resistors), GPS and LoRa, streaming live readings over Wi-Fi to a web dashboard.
- Led the technical research and helped build the LoRa link for the Damietta Hackathon.

[Robo-Space](#) — Predictive Maintenance with Machine Learning

Jun 2026

- Random Forest classifier detecting 6 machine-failure types from temperature, RPM and torque, with a Tkinter dashboard, ROC and feature-importance plots, and text-to-speech alerts (team of 6).

[NumLab](#) — Numerical Methods Solver

Jun 2026

- Desktop solver with every numerical algorithm hand-written in Python and a CustomTkinter interface (team of 6).

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, GDScript, Supabase

AI & Data: scikit-learn, Pandas, Matplotlib, Random Forest, Gemini API, Computer Vision (YOLO)

Hardware & IoT: ESP32 (XIAO ESP32-C6), Arduino IDE, LoRa, MPU6050, MAX30100, GPS modules

Games & 3D: Godot 4, Blender, game balancing, multi-platform export

Web & Tools: React, Next.js, Vite, Tailwind CSS, CustomTkinter, Git & GitHub, GitHub Actions

LEADERSHIP & COMMUNITY

- Competitive Programmer, [ICPC HUE](#) — training with the community's 16 teams on algorithms and data structures; team ranked #1 at Horus in the ECPC 2026 qualifications. Jan 2026 – Present
- Member of Technical Staff, [AXIS](#) — technical member of the student club at the Faculty of Artificial Intelligence, Horus University. Dec 2025 – Present